

Pinkeye Detection and Prediction

Iman Khazrak^a, Rob Green^a, Pei Wang^b, Jameson Brennan^c, Hossein Moradi Rekabdarkolae^b

^a*Department of Computer Science, Bowling Green State University, Bowling Green, Ohio, USA*

^b*Department of Business Analytics, Economics, and Information Systems, Bowling Green State University, Bowling Green, Ohio, USA*

^c*Department of Animal Science, South Dakota State University West River Research and Extension, Rapid City, South Dakota, USA*

Abstract

Infectious bovine keratoconjunctivitis (IBK), commonly known as pinkeye, is a highly contagious ocular disease in cattle that causes significant welfare and economic losses. Early automated detection from farm images can enable timely veterinary intervention. We present a two-stage deep learning pipeline: *Stage 1* localizes the eye region using a YOLOv8 detector, and *Stage 2* classifies the cropped eye as healthy or diseased using four architectures: ResNet18, EfficientNet-B0, a baseline CNN+Transformer hybrid, and an enhanced CNN+Transformer variant with a class token and deeper head. Evaluated on 394 labeled eye crops (309 healthy, 85 pinkeye) under stratified 5-fold cross-validation, the Strong CNN+Transformer achieves the highest mean accuracy (95.94 %) and the highest mean F1-score on the minority pinkeye class (90.51 %); ResNet18 attains the highest mean ROC-AUC (98.67 %) among the four models in the primary benchmark. Augmentation ablations show architecture-dependent gains and trade-offs. These results support eye-localized classification as a feasible screen for this imbalanced, fine-grained imaging setting.

Keywords: Pinkeye detection, YOLOv8, CNN+Transformer, class imbalance, data augmentation, cattle health monitoring, transfer learning.

1. Introduction

Bovine pinkeye, also called infectious bovine keratoconjunctivitis (IBK), is a very contagious eye disease that occurs in cattle around the world. It is

widely known as the most common and important eye problem in both beef and dairy cattle systems (Brown et al., 1998). The disease is mainly caused by bacteria such as *Moraxella bovis*. Common signs include watery eyes, swelling of the conjunctiva, ulcers on the cornea, and in more serious cases, damage to the cornea and even blindness (Kizilkaya et al., 2013). Although IBK does not usually cause death, it can strongly affect animal well-being and farm productivity, mostly because infected cattle eat less and have trouble seeing (Kizilkaya et al., 2013).

In the United States, IBK causes serious financial losses for the cattle industry. It is viewed as one of the costliest diseases in beef production, with yearly losses estimated to be more than 150 million. These losses come from slower growth, treatment expenses, and extra management costs (Bartenslager et al., 2021; Angelos, 2015). One major reason for the loss is lower weight gain in infected animals. For instance, calves with IBK can have weaning weights about 8.9 kg less than healthy calves (Kilama et al., 2025). Besides weight loss, the disease can also reduce milk yield and overall performance. This happens because infected cattle experience pain and stress, and they do not use feed as efficiently (Kilama et al., 2025).

IBK continues to be a serious problem for the cattle industry because it occurs often, has many possible causes, and leads to large economic losses. For these reasons, researchers are still working on better ways to detect the disease early, prevent it, and manage it more effectively. The goal of this work is to reduce production losses and improve the health and well-being of cattle (Gupta et al., 2023).

Methods used to identify IBK have changed over time, moving from simple visual checks to more advanced molecular and data-based techniques. In the past, diagnosis mainly depended on visible signs such as tearing, eye inflammation, and cloudy corneas. However, these signs usually show up after the disease has already started, and they can look similar to other eye problems, which makes accurate diagnosis more difficult (Snowder et al., 2005; Angelos, 2015). Because of these limits, more recent studies have focused on finding earlier and more specific ways to detect the disease, especially methods that target the exact pathogen.

Modern laboratory methods have improved IBK detection by directly identifying causative bacteria. Polymerase chain reaction (PCR) is widely used to detect *Moraxella bovis* and *Moraxella bovoculi*, offering higher sensitivity and accuracy than traditional culture methods (Angelos et al., 2007; Angelos, 2015). Matrix-assisted laser desorption/ionization time-of-flight

mass spectrometry (MALDI-TOF MS) also enables rapid and reliable differentiation of closely related bacterial species (Seng et al., 2009). More recently, next-generation sequencing (NGS) has been applied to study the bovine ocular microbiome, showing that IBK is linked to shifts in microbial communities rather than a single pathogen, supporting its multi-factorial nature (Holman et al., 2017).

Because molecular tests are often costly and take time to complete, recent studies have explored data-driven methods for earlier IBK detection. In particular, machine learning and image-based approaches can analyze eye images and animal behavior to identify early signs of the disease. These methods are non-invasive and can support real-time monitoring (Gupta et al., 2023; Mahmud et al., 2021). These studies reflect a shift toward automated and predictive systems that can help farmers monitor large herds more efficiently and respond to disease at an earlier stage (Mahmud et al., 2021; Mia et al., 2025).

In this study, we propose a two-step framework for detecting pinkeye in cattle. The first step focuses on localizing and cropping the eye region, while the second step benchmarks classification performance. To address data imbalance, several data augmentation strategies are evaluated. In the classification stage, multiple state-of-the-art deep learning models are applied and compared. Overall, the eye-only approach demonstrates strong potential for pinkeye screening, with both model selection and augmentation strategies having a notable impact on performance for the minority class.

2. Data

All animal work followed established ethical guidelines for the use of animals in research and was carried out under an approved protocol from the South Dakota State University Institutional Animal Care and Use Committee (IACUC; protocol #2501-010E). The study took place at the South Dakota State University Cottonwood Field Station (CFS) in the mixed-grass prairie of the Northern Great Plains, where elevation ranges from 710 to 784 m. Data were collected during the summers of 2023–2024 using yearling Angus beef steers, which were managed through the CFS livestock handling facilities and processed in a Silencer™ hydraulic squeeze chute (Moly mfg., Lorraine, KS, USA). For each steer, two lateral (left and right) images were acquired with a digital camera during handling. Animals diagnosed with pink eye received treatment according to the standard health management procedures

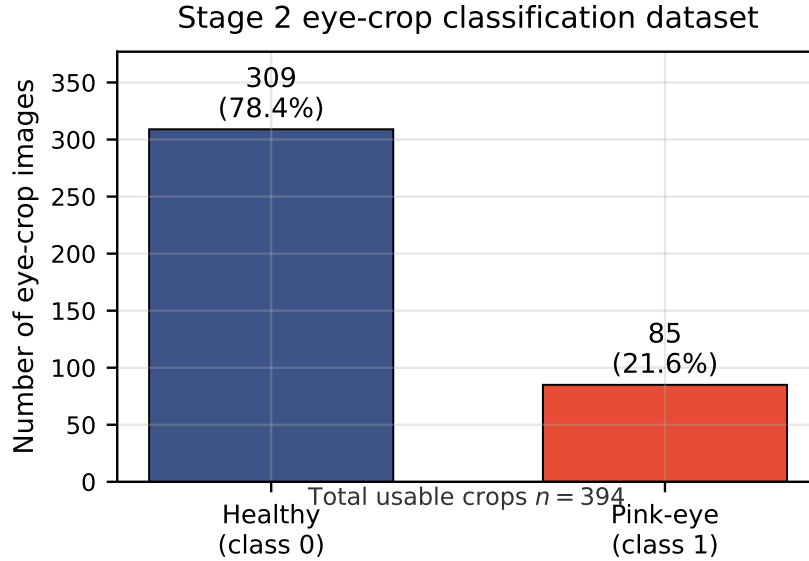


Figure 1: Class distribution of usable Stage 2 eye crops ($n=394$): 309 healthy (78.4 %) and 85 pinkeye (21.6 %), matching folder labels after YOLO cropping.

in place at the Cottonwood Field Station. All photographs used in this work were obtained with a FujiFilm FinePix AX655 16MP digital camera equipped with a CMOS sensor, 5x optical zoom, and optical plus digital image stabilization, producing high-resolution images suitable for subsequent analysis. Image files were standardized to JPG format and then manually annotated by trained personnel as either healthy or infected. There are a total of 408 JPG images as final dataset. These images were manually classified into two categories: healthy (323 images) and defective (85 images), and stored in separate folders accordingly. Figure 1 summarizes the Stage 2 class counts used in all experiments.

3. Methods

The proposed system is a two-stage cascaded pipeline: stage 1 detect the eye region in each cattle head image and stage 2 classify the cropped eye images to discriminate between healthy eyes and eyes affected by IBK. This approach is detailed as follows.

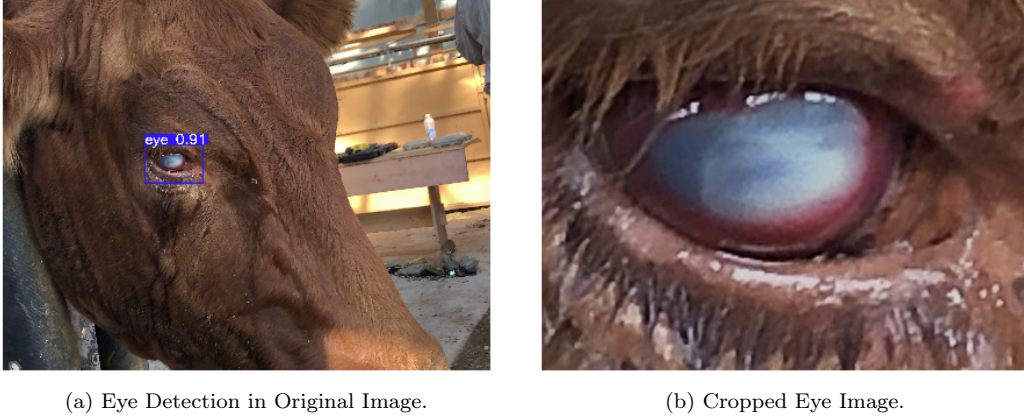


Figure 2: Eye Localization and Crop

3.1. Stage 1: Eye Localization with YOLOv8

Since the goal is to predict if the eye is infected or not, the first logical step is to find and localize the eye. You Only Look Once (YOLO; Redmon et al., 2016) is a widely used algorithm for fast and accurate objective detection approach. We employed YOLOv8 (Yaseen, 2024) detector and trained it on 100 cattle head images sourced from the Roboflow Eye Detection dataset. Each image carries a single bounding-box annotation for the `eye` class in YOLO normalized format. All 100 annotation files were validated (pairing, coordinate ranges, class ID = 0) prior to training. The dataset was split deterministically with seed 42 into training (80 %, 93 images), validation (10 %, 11 images), and test (10 %, 13 images), using the same stratification rule as in the logged detector training run.

The best detector checkpoint was applied to all 408 classification images with confidence threshold 0.25 and symmetric padding 10 % around each box (clamped to image bounds), matching the detector inference settings above (confidence 0.25, 10 % symmetric padding). Figure 2 illustrates a representative example of the eye detection and subsequent cropping process. The Roboflow export was checked with a validation script (format, bounds, class ID). All 100 labels are passed to the stage 2.

A YOLOv8n model pretrained on MS-COCO was fine-tuned on the 93-image training split using the Ultralytics YOLOv8 API (Jocher et al., 2023). The hyperparameters match the logged training run in the project detector report (Table 1).

Table 1: YOLOv8n Training Hyperparameters (Stage 1)

Parameter	Value
Base model	YOLOv8n (pretrained, MS-COCO)
Epochs	100
Image size	640×640
Batch size	16
Optimizer	AdamW (framework default LR schedule)
Training images	93
Validation images	11
Test images	13
Random seed	42

3.2. Stage 2: Classification Models

The classification pool comprised 408 cattle head images with folder labels (healthy/pinkeye). After YOLOv8 inference and cropping at confidence 0.25, 394 images yielded usable eye crops; 14 images had no passing detection and were set aside for manual review. The Stage 2 dataset therefore contains 309 healthy and 85 pinkeye crops (78.4% / 21.6%). Images are resized to 224×224 for classification. The following deep learning models are used to classify the images:

- **ResNet18** (He et al., 2016) with ImageNet-1k weights was adapted by replacing the final fully connected layer with a $512 \rightarrow 2$ head and fine-tuning the full network.
- **EfficientNet-B0** (Tan and Le, 2019) with ImageNet-1k weights was adapted with a $1280 \rightarrow 2$ classifier head and end-to-end fine-tuning.
- **Baseline CNN+Transformer Hybrid:** The baseline hybrid feeds final-stage CNN feature maps through a 1×1 projection, flattens spatial locations into tokens, applies a Transformer encoder, mean-pools tokens, and classifies with a linear head in our implementation.
- **Strong CNN+Transformer:** The Strong variant uses an EfficientNet-B0 backbone (features-only), 1×1 projection with BatchNorm and GELU, patch tokens with a learnable CLS token and learnable positional embeddings, a pre-norm Transformer encoder ($L=3$, $h=4$, $d=256$),

Table 2: Strong CNN+Transformer Architectural Hyperparameters

Parameter	Value
CNN backbone	EfficientNet-B0 (pretrained)
Embedding dimension d	256
Transformer heads h	4
Transformer layers L	3
MLP ratio	4.0
Dropout	0.1
Attention dropout	0.1
Token norm	Pre-LayerNorm
Activation	GELU

and an MLP classification head on the CLS output (Wightman). Hyperparameters are summarized in Table 2.

For the stage 2, with 394 crops, we used stratified 5-fold cross-validation (seed 42), preserving the healthy/pink-eye ratio in each fold. Each fold uses four parts for training and one for validation; metrics are summarized as mean and standard deviation across folds. Training used weighted cross-entropy with class weights $w_c = N/(Kn_c)$ over the *training-fold* sample counts (inverse frequency within each fold). This upweights the minority pinkeye class relative to healthy samples. Optimization used AdamW (Loshchilov and Hutter, 2017) with learning rate 10^{-4} and weight decay 10^{-4} . Early stopping monitored validation F1 on the pinkeye class with patience 8 and minimum improvement $\delta = 10^{-4}$. For each validation fold we report accuracy, balanced accuracy, precision/recall/F1 for the pinkeye (minority) class, and ROC-AUC (one-vs-rest using predicted pinkeye probability). The primary score for comparing augmentation settings is F1 on pinkeye denoted by $F1_{PE}$.

All Step 2 benchmarks and augmentation studies were run as batch GPU jobs on a shared cluster managed by the SLURM workload scheduler. Every study used the same stratified 5-fold split (seed 42), weighted cross-entropy, AdamW learning rate 10^{-4} (weight decay 10^{-4}), batch size 8, early stopping on validation $F1_{PE}$ with patience 8 (minimum improvement $\delta=10^{-4}$), and CUDA acceleration with the centre’s module stack. Training was capped at five epochs per fold unless stopped earlier by early stopping. Default settings used for ad hoc interactive runs may differ; all figures and tables in Section 4

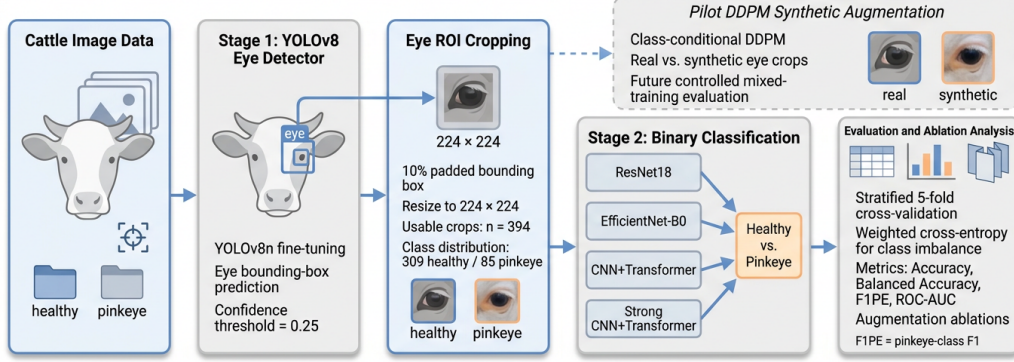


Figure 3: Overall workflow of the proposed two-stage bovine pinkeye detection pipeline.

follow this cluster protocol.

The proposed two-stage cascaded pipeline, illustrated in Figure 3. In the first stage a YOLOv8-based object detector localizes the eye region in each cattle head image. The detected bounding box is padded and cropped, producing a standardized eye-region image that forms the input to Stage 2. In the second stage, four deep classification models are trained and evaluated on the cropped eye images to discriminate between healthy eyes and eyes affected by IBK.

4. Results

In the first step of eye localization, a YOLOv8-based detector was developed using a manually annotated dataset of 117 cow images with a single target class (eye), for which all annotations were verified as valid with no errors. The dataset was divided into training, validation, and test sets of 93, 11, and 13 images, respectively. Model performance was evaluated using standard detection metrics, including precision, recall, mAP50 (mean Average Precision at a 0.50 intersection over union threshold), and mAP50-95 (averaged across stricter intersection over union thresholds). The detector achieved near-perfect results (see Table 3), with a precision of 1.000, recall of 0.999, and mAP@50 of 0.995, while the mAP50-95 score of 0.631 indicates strong localization performance under more stringent criteria.

Table 3: Eye Detection Accuracy

Metric	Precision	Recall	mAP50	mAP50-95
Value	1.000	0.999	0.995	0.631

To assess its applicability for downstream classification, the trained detector was applied to a sample dataset of 408 images, successfully identifying eye regions in 394 images, with only 14 images lacking detections and retained for manual review. The detected eye regions were subsequently cropped, resulting in a classification dataset of 394 images, including 309 healthy cases and 85 pinkeye cases. These results demonstrate that the proposed eye localization approach is highly reliable and provides a strong foundation for the subsequent pinkeye classification task.

Following the eye localization and cropping process, the resulting classification dataset reflects a moderate class imbalance. Four following training augmentations were compared.

1. **Full augmentation** defines the primary benchmark: random resized crop, horizontal flip, rotation ($\pm 10^\circ$), color jitter, and mild affine transforms, with validation and test limited to resize and ImageNet normalization Wightman.
2. **No-color augmentation** retained flip, rotation, and light Gaussian noise but omitted color jitter.
3. **Flip and rotation only** removed Gaussian noise relative to (ii).
4. **Minority-only flip and rotation** applied geometric augmentation only to pinkeye-labeled training crops.

We compared these four augmentation policies keeping the same folds, loss, and stopping rule within each compared study. Model sets varied slightly by experiment (all four models for configurations (1) and (2); subsets for (3) and (4) as listed in the tables). Figure 4 shows an example for healthy and Pinkeye crops and class-conditional DDPM samples from the pilot synthetic augmentation study.

To rigorously evaluate model performance, a 5-fold stratified cross-validation (CV) framework was employed to preserve class proportions across folds. Each model was trained for up to 5 epochs per fold, with early stopping implemented to prevent overfitting. To further address class imbalance, a weighted cross-entropy loss function was adopted, assigning higher importance to the minority (pinkeye) class. This evaluation protocol ensures a

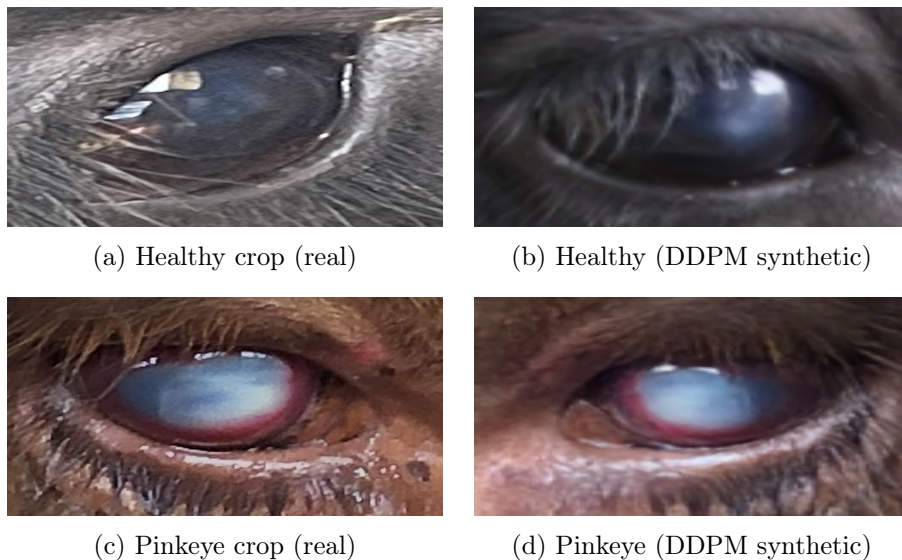


Figure 4: Qualitative comparison of real Stage 2 crops and class-conditional DDPM samples from the pilot synthetic augmentation study. The top row shows healthy examples, and the bottom row shows pinkeye examples. For each class, a real cropped eye image is shown next to a synthetic image generated by the corresponding class-conditional DDPM.

robust and fair comparison of classification models while promoting stable and generalizable performance. Table 4 lists 5-fold mean and the standard deviation of different metrics for different models under the full augmentation configuration. Figure 5 visualizes mean $F1_{PE}$ with error bars using the same fold-wise summaries as Table 4.

Overall, all models demonstrate strong classification performance, with accuracy values above 0.92 and AUCs exceeding 0.96, indicating good discriminative ability across methods. Among them, Strong CNN+Transformer achieves the best overall performance, with the highest accuracy (0.9594) and F1-score (0.9051), suggesting improved balance between precision and recall compared to other models. Its AUC (0.9850) is also competitive, indicating stable ranking performance. ResNet18 and EfficientNet-B0 perform similarly, with nearly identical accuracy (around 0.957). However, EfficientNet-B0 slightly outperforms ResNet18 in balanced accuracy (0.9383 vs. 0.9342) and recall (0.9059 vs. 0.8941), suggesting better handling of class imbalance and improved sensitivity. The baseline CNN+Transformer model underperforms relative to others across most metrics, particularly in F1-score (0.8369) and accuracy (0.9239), indicating that without further optimization, hybrid

Table 4: Stage 2 classification under full augmentation. Result are the average and standard deviation in parenthesis of the 5-fold. Best per column in bold.

Model	Accuracy	Bal. Acc.	Precision _{PE}	Recall _{PE}	F1 _{PE}	ROC-AUC
ResNet18	0.9569 (0.0144)	0.9342 (0.0289)	0.9094 (0.0619)	0.8941 (0.0644)	0.8992 (0.0342)	0.9867 (0.0117)
EfficientNet-B0	0.9568 (0.0114)	0.9383 (0.0297)	0.8980 (0.0393)	0.9059 (0.0671)	0.9000 (0.0279)	0.9829 (0.0122)
CNN+	0.9239 (0.0235)	0.9132 (0.0215)	0.7934 (0.0928)	0.8941 (0.0492)	0.8369 (0.0429)	0.9681 (0.0327)
Transformer						
Strong CNN+	0.9594	0.9358	0.9251	0.8941	0.9051	0.9850
Transformer	(0.0106)	(0.0229)	(0.0795)	(0.0644)	(0.0214)	(0.0024)

Table 5: Per-fold accuracy and F1_{PE} (full augmentation).

Fold	ResNet18		Strong CNN+T	
	Acc.	F1 _{PE}	Acc.	F1 _{PE}
1	0.9494	0.8889	0.9494	0.8889
2	0.9620	0.9143	0.9747	0.9412
3	0.9747	0.9412	0.9494	0.8889
4	0.9367	0.8485	0.9620	0.9032
5	0.9615	0.9032	0.9615	0.9032
Mean	0.9569	0.8992	0.9594	0.9051
Std	0.0144	0.0342	0.0106	0.0214

architectures may not fully leverage their potential. In terms of variability, all models exhibit relatively small standard deviations, indicating stable performance across runs, with Strong CNN+Transformer showing particularly consistent AUC (± 0.0024). Table 5 gives per-fold accuracy and F1_{PE} for ResNet18 and Strong CNN+Transformer. Table 6 aggregates confusion counts across folds for Strong CNN+Transformer. Fold-level confusion matrices, ROC curves, and learning curves were saved with the original GPU experiments but are omitted here for brevity.

We further analyze the dataset with augmentation experiments to deal with the imbalance issue. Three different augmentations have been included: with color-based augmentation, Gaussian noise augmentation, and minority-

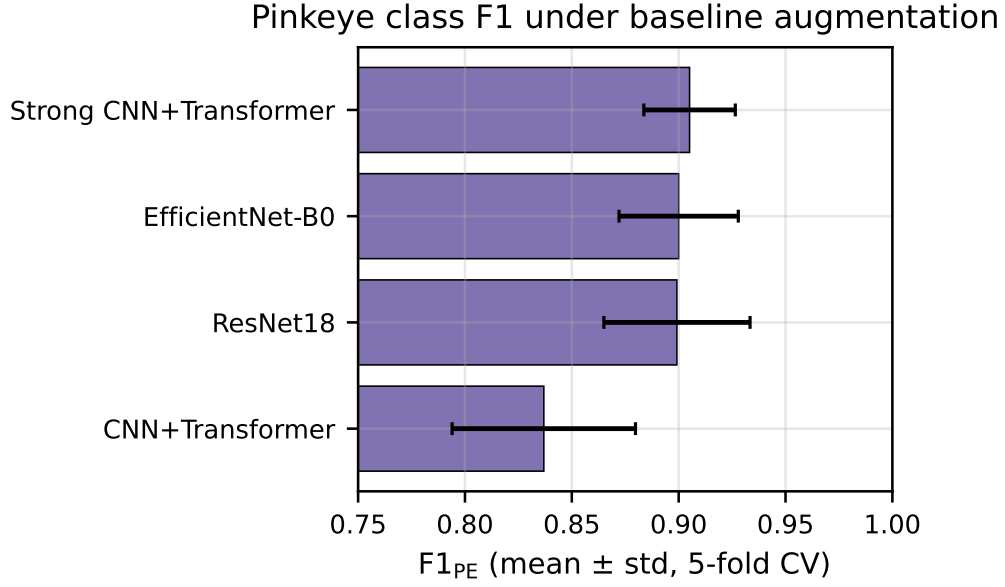


Figure 5: Mean $\pm$ std $F1_{PE}$ under full augmentation, replotted from the same statistics as Table 4.

Table 6: Aggregated confusion matrix for Strong CNN+Transformer (sum over validation folds; total samples = 394).

True \ Pred	Healthy	Pink-eye
Healthy	302	7
Pink-eye	9	76

Table 7: Effect of removing color-related augmentation: mean $F1_{PE}$ and accuracy (5-fold mean).

Model	$F1_{PE}$ (full)	$F1_{PE}$ (no-col)	$\Delta F1$	ΔAcc
ResNet18	0.8992	0.9190	+0.0198	+0.0076
EfficientNet-B0	0.9000	0.8721	−0.0279	−0.0152
CNN+Transformer	0.8369	0.8272	−0.0097	−0.0001
Strong CNN+T	0.9051	0.8672	−0.0379	−0.0153

only augmentation. We first evaluated the impact of color-based augmentation by comparing a full augmentation strategy (including color jitter and affine transformations) with a reduced “no-color” setup, which applied only geometric transformations (flipping, rotation) and mild Gaussian noise. The results reveal model-dependent effects. For ResNet18, removing color and affine augmentations led to improved performance, with increases in F1-score (+0.0198) and accuracy (+0.0076). In contrast, EfficientNet-B0 and the Strong CNN+Transformer experienced performance declines under the no-color setting, with decreases in F1-score of −0.0279 and −0.0379, respectively. These findings suggest that color and affine augmentations contribute positively to performance for more complex architectures, likely by enhancing feature diversity and robustness. However, for ResNet18, a simpler augmentation strategy appears more effective, potentially due to reduced over-regularization or better alignment with the model’s capacity.

To assess the contribution of Gaussian noise, we compared a setting using only geometric transformations (flip and rotation) with the no-color augmentation setup, which additionally included Gaussian noise. The results indicate that Gaussian noise has differing effects across model architectures. For ResNet18 and EfficientNet-B0, removing Gaussian noise led to decreases in F1-score (−0.0210 and −0.0202, respectively), suggesting that noise injection serves as an effective regularization mechanism for these CNN-based models. In contrast, the Strong CNN+Transformer model showed an improvement in F1-score (+0.0117) when Gaussian noise was excluded, indicating that such perturbations may interfere with the hybrid model’s feature learning process. Overall, these findings suggest that Gaussian noise augmentation benefits standard CNN architectures by improving robustness, but may be detrimental to more complex hybrid models, where it could introduce unnecessary variability.

Table 8: Gaussian noise ablation: mean $F1_{PE}$. Δ is flip+rotation (no noise) minus no-colour+noise.

Model	No-col+noise	Flip+rot	$\Delta F1_{PE}$
ResNet18	0.9190	0.8979	-0.0210
EfficientNet-B0	0.8721	0.8518	-0.0202
Strong CNN+T	0.8672	0.8788	+0.0117

Table 9: Minority-only versus all-class geometric augmentation (mean $F1_{PE}$).

Model	No-col+noise	Flip+rot (all)	Flip+rot (min.)	Best setting
ResNet18	0.9190	0.8979	0.9060	no-col+noise
EfficientNet-B0	0.8721	0.8518	0.8624	no-col+noise
Strong CNN+T	0.8672	0.8788	0.8582	flip+rot (all)

We further evaluated a targeted augmentation strategy in which geometric transformations (flip and rotation) were applied exclusively to the minority class (pink eye). Under this setting, ResNet18 achieved the best performance among the evaluated models, with an F1-score of $0.9060(\pm 0.0282)$. However, this configuration did not surpass the best results obtained under previous augmentation strategies for any of the models. These findings suggest that while minority-only augmentation can provide modest improvements in specific cases, it does not consistently enhance overall model performance. In particular, focusing augmentation solely on the minority class may be insufficient to fully address class imbalance, indicating that more comprehensive or balanced augmentation strategies are likely needed.

To summarize, the results demonstrate that eye-crop-based classification is a viable approach, as consistently strong performance across multiple architectures indicates that the eye region contains sufficient discriminative information for the task. Model performance, however, is sensitive to the choice of augmentation strategy. While the Strong CNN+Transformer achieves the best overall performance under the baseline setting, ResNet18 exhibits greater robustness under more constrained or controlled augmentation regimes. The use of weighted cross-entropy in conjunction with stratified cross-validation effectively mitigates class imbalance, yielding stable minority-class recall and F1-scores across models. Notably, the augmentation policy emerges as a critical factor influencing performance, with even minor changes in transformation strategies leading to substantial variations

Table 10: Best mean accuracy / $F1_{PE}$ by augmentation regime (5-fold means).

Configuration	Best model	Accuracy	$F1_{PE}$
Full augmentation (color + affine)	Strong CNN+Transformer	0.9594 0.0106	0.9051 0.0214
No color (+ Gaussian noise)	ResNet18	0.9645 0.0165	0.9190 0.0374
Geometric only (flip + rotation)	ResNet18	0.9543 0.0147	0.8979 0.0312
Minority-only augmentation	ResNet18	0.9594 0.0105	0.9060 0.0282

in minority-class F1-score.

5. Discussion

Strong CNN+Transformer leads on mean accuracy and $F1_{PE}$ when the richer augmentation policy is used, while ResNet18 achieves the best mean ROC-AUC in Table 4 and the best mean $F1_{PE}$ under the no-color regime (Table 10). The ablations indicate that color and affine regularization in policy (i) particularly benefit the deeper CNN and hybrid models, whereas the shallow ResNet benefits when color perturbations are omitted.

Gradient-based Grad-CAM visualizations Selvaraju et al. (2017) were computed for the CNN baselines during training. Illustrative overlays are not reproduced in this manuscript; qualitatively, saliency tended to align with conjunctival and peripheral corneal regions on pinkeye crops, consistent with clinical presentation.

The dataset size (85 pinkeye crops) limits generalization claims. A class-conditional DDPM (Ho et al., 2020) for synthetic augmentation was trained to address this challenge. Systematic Stage 2 fine-tuning with curated synthetic augmentation and reporting against the benchmark protocol remains future work. Broader directions include larger multi-site collections, severity grading, and external validation. An additional possible augmentation strategy for this study is diffusion-based synthetic data generation using a denoising diffusion probabilistic model (DDPM). Specifically, class-conditional DDPM models were trained to generate synthetic eye images for both healthy and pink eye classes. The training was conducted using a batch size of 16,

10,000 diffusion time steps, and 2,048 epochs. The primary objective of this extension is to enhance model robustness and address class imbalance by augmenting the limited number of real pink eye samples with high-quality synthetic data.

6. Conclusion

In this study, we advance early detection of infectious bovine keratoconjunctivitis (IBK) by evaluating deep learning classifiers on curated ocular image crops and exploring augmentation policies that enhance model robustness. Building on the need for earlier, field-deployable screening beyond clinical signs and post-onset diagnostics (Brown et al., 1998; Angelos, 2015) (Fenwick and Rider, 1998; Angelos, 2015), our results indicate that carefully designed color and affine regularization particularly benefit deeper CNN and hybrid architectures, while shallower ResNet variants perform best when color perturbations are reduced. Grad-CAM saliency localized to conjunctival and peripheral corneal regions on positive cases, aligning model attention with expected clinical pathology and providing face-valid interpretability for chute-side or pasture-based screening workflows.

Despite promising findings, the dataset size (85 pinkeye crops) constrains generalizability and motivates larger, multi-site collections with standardized imaging, breed diversity, and varied lighting. To address data scarcity, we introduced a class-conditional denoising diffusion probabilistic model (DDPM) to generate synthetic ocular images as a principled path toward augmentation, with systematic Stage 2 fine-tuning and benchmarking defined as future work (Ho et al., 2020). Downstream priorities include prospective external validation, severity grading, and comparisons against skilled human raters; evaluation of object-detection pipelines (e.g., Ultralytics YOLO) for end-to-end eye localization in unconstrained settings; and integration with laboratory and molecular tools such as PCR for *Moraxella* spp. to form hybrid decision-support systems (Angelos et al., 2007; Angelos, 2015). Longer term, coupling phenotypic signals with genomic and microbiome insights may enable proactive risk stratification at the herd level (Kizilkaya et al., 2013; Holman et al., 2017; Gupta et al., 2023).

Collectively, this work demonstrates that modern computer vision methods supported by interpretable attention maps and rigorous augmentation can help shift IBK management from reactive treatment to earlier, targeted intervention. Such tools have the potential to reduce production losses, im-

prove animal welfare, and support judicious antimicrobial use, provided they are validated across environments and integrated into practical husbandry workflows. By outlining a reproducible benchmark and a roadmap for synthetic augmentation and external validation, we establish a foundation for scalable, field-ready IBK screening systems that complement laboratory diagnostics and accelerate data-driven cattle health management.

7. Disclosure statement

The authors declare no relevant disclosures of competing interests.

8. Data Availability Statement

This data and codes for this work will become publicly available upon acceptance.

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